Mass Gap for the O(N) Model: Analysis of the HS Approach

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April 2026

Abstract

We analyze the Hubbard-Stratonovich approach to proving the mass gap for the O(N) model in two dimensions at large N. The HS identity for the Euclidean quartic requires imaginary coupling, creating a sign problem. We compare the NLSM (where Kupiainen's constraint trick gives a real mass m_0^2 in the operator, enabling a direct FK bound for each σ-value) with the LSM (where the real mass cancels after HS, and the mass gap must emerge from the oscillatory σ-average). A comparison across $d = 0, 1, 2$ shows that the sign problem is specific to $d = 2$: the logarithmic UV divergence of the Green's function promotes the self-energy from $O(1/N)$ to $O(1)$, causing dimensional transmutation. We propose a proof strategy based on the **Lefschetz thimble**: a contour deformation to the steepest descent manifold through the dominant saddle, where the integrand becomes real, positive, and approximately log-concave. At large N, a single thimble dominates (exponential suppression of competitors), eliminating the sign problem and enabling Brascamp-Lieb concentration and Feynman-Kac bounds. We compare four strategies and discuss the current status of the Lean formalization.

Contents

1 The HS identity and the sign problem

1.1 The Euclidean quartic requires imaginary coupling

The O(N) LSM action on $\Lambda \subset \mathbb{Z}^2$:

$$S[\varphi] = \tfrac{1}{2}\sum_i \langle \varphi^i, (-\Delta)\varphi^i \rangle + N \sum_x \lambda\big(\tfrac{|\varphi(x)|^2}{N} - \rho^2\big)^2. \tag{1}$$

The Boltzmann weight is e^{-S}. The quartic appears as $e^{-\lambda a^2}$ with $a = |\varphi|^2/N - \rho^2$. The HS identity:

$$e^{-\lambda a^2} = \frac{1}{\sqrt{4\pi\lambda}} \int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda) + i\sigma a}\, d\sigma. \tag{2}$$

Here $\sigma \in \mathbb{R}$ (real integration variable), $e^{-\sigma^2/(4\lambda)}$ ensures convergence, and the coupling $i\sigma a$ is **imaginary**. This is a Fourier transform (proved from Mathlib's `fourierIntegral_gaussian`).

The imaginary coupling is forced by the *minus* sign in $e^{-\lambda a^2}$ (repulsive/confining quartic in Euclidean signature). A real coupling would give $e^{+\lambda a^2}$ (attractive, wrong sign).

1.2 The σ-integral on the real axis

After applying (2) at each site and integrating out φ (Gaussian with complex mass), the partition function becomes:

$$Z = c \int_{\mathbb{R}^{|\Lambda|}} \det\bigl(-\Delta + 2i\sigma z\bigr)^{-N/2} \, e^{-\sum_x [\sigma(x)^2/(4\lambda) + i\sigma(x)\rho^2]} \, d^{|\Lambda|}\sigma, \tag{3}$$

where $z = N^{-1/2}$ and the det comes from the φ-integral.

This integral **converges** on the real σ-axis:

- $e^{-\sigma^2/(4\lambda)}$ provides Gaussian decay in σ.
- $|\det(-\Delta + 2i\sigma z)^{-N/2}| \le \det(-\Delta)^{-N/2}$ (bounded, since $|1 + i\theta| \ge 1$ for real θ).

But the integrand is **complex** (the det and the $e^{i\sigma\rho^2}$ factor oscillate). This is the sign problem.

2 The NLSM: Kupiainen's mass trick

2.1 Adding m_0^2 for free

For the NLSM with constraint $|\varphi|^2 = N\beta$: any function of $|\varphi|^2$ is constant on the constraint surface. In particular, $\frac{1}{2} m_0^2 |\varphi|^2 = \frac{1}{2} m_0^2 N\beta = \text{const}$.

So we can **freely add** $\frac{1}{2} m_0^2 |\varphi|^2$ to the action, giving the massive GFF:

$$S_{\text{NLSM}} = \tfrac{1}{2}\langle \varphi, (-\Delta + m_0^2)\varphi\rangle + \text{const}, \qquad |\varphi|^2 = N\beta. \tag{4}$$

The parameter $m_0^2 > 0$ is arbitrary; we choose it via the gap equation (Kupiainen eq. 9b): $(-\Delta + m_0^2)^{-1}_{ii} = \beta$.

2.2 The NLSM operator has real mass

After HS (introducing the Lagrange multiplier α for the constraint), the φ-propagator at fixed α is:

$$\bigl(-\Delta + m_0^2 - 2i\alpha z\bigr)^{-1}(x, 0). \tag{5}$$

The operator has **real part** $-\Delta + m_0^2 \ge m_0^2 > 0$. By the triangle inequality (FK representation):

$$\bigl|(-\Delta + m_0^2 - 2i\alpha z)^{-1}(x, 0)\bigr| \le (-\Delta + m_0^2)^{-1}(x, 0) \sim e^{-m_0|x|}. \tag{6}$$

Every α-value gives exponential decay with mass m_0. The α-average is a correction, not the leading effect.

2.3 Kupiainen's proof structure

1. Gap equation: $m_0 > 0$ (always in 2d).
2. Bound (6): each α-value gives decay.
3. Cluster expansion + RP: control the α-average (prefactor, not the mass).
4. Mass gap: $m = m_0(1 + O(1/N))$.

The hard step is (3), but the mass m_0 comes for free from (2).

3 The LSM: the mass cancels

3.1 Add-subtract fails

For the LSM (no constraint), we try the same trick: add $\frac{1}{2}m_0^2|\varphi|^2$ to the action and subtract it from the interaction:

$$S = \left[\tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle\right] + \left[\lambda\left(\tfrac{|\varphi|^2}{N} - \rho^2\right)^2 - \tfrac{1}{2}m_0^2\,\tfrac{|\varphi|^2}{N}\right]. \tag{7}$$

Apply HS to $\lambda(|\varphi|^2/N - \rho^2)^2$ only. The subtracted mass $-\frac{1}{2}m_0^2\,|\varphi|^2/N$ remains outside the HS. Computing the φ-dependence:

$$\begin{aligned} &-\tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle + i\sigma|\varphi|^2 + \tfrac{1}{2}m_0^2|\varphi|^2 \\ &\quad = -\tfrac{1}{2}\langle\varphi, (-\Delta)\varphi\rangle + i\sigma|\varphi|^2 \\ &\quad = -\tfrac{1}{2}\langle\varphi, (-\Delta - 2i\sigma)\varphi\rangle. \end{aligned} \tag{8}$$

The m_0^2 from the GFF and the $-m_0^2$ from the subtraction **cancel exactly**. The LSM operator is:

$$-\Delta - 2i\sigma \qquad (\textbf{no real mass}). \tag{9}$$

3.2 Consequences

The operator (9) has eigenvalues $k^2 - 2i\sigma$ with real part $k^2 \geq 0$ (from $-\Delta$) and imaginary part -2σ. For $k = 0$: the eigenvalue is $-2i\sigma$ (purely imaginary).

- **No single-σ decay**: $|(-\Delta - 2i\sigma)^{-1}(x, 0)|$ does NOT decay exponentially for each σ-value. At $k = 0$: $|1/(-2i\sigma)| = 1/(2|\sigma|)$, independent of $|x|$.
- **Mass from σ-average only**: The exponential decay of $\langle\varphi(x)\varphi(0)\rangle$ must come *entirely* from the oscillatory cancellations in the σ-integral (3).
- **The FK bound fails**: The NLSM bound (6) relies on the real mass m_0^2. Without it, $|G_\sigma| \leq G_0$ gives $G_0 = (-\Delta)^{-1}$ (massless propagator, no decay).

3.3 Comparison

	NLSM	LSM
Mass in operator	m_0^2 (free, from constraint)	**none** (cancels)
Operator after HS	$-\Delta + m_0^2 - 2i\alpha z$	$-\Delta - 2i\sigma$
Single-σ decay	$e^{-m_0\|x\|}$ (yes)	**none**
Mass gap source	Real mass + corrections	**σ-average only**
FK bound	Works ($\|G\| \le G_{m_0}$)	**Fails**

4 Comparison across dimensions

The sign problem and mass gap structure depend strongly on dimension.

4.1 $d = 0$ (matrix model)

One site ($|\Lambda| = 1$). The σ-integral is one-dimensional: $\int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda)+i\sigma a}\, d\sigma$. The saddle-point Gaussian is exact up to $O(1/N)$ corrections. Oscillations are $O(1/N^2)$ — negligible. No sign problem.

The Green's function $C = 1/m_0^2$ is a number. The self-energy $\Sigma = O(1/N)$. The mass is perturbative: $m = m_{\text{bare}}(1 + O(1/N))$. No dimensional transmutation.

4.2 $d = 1$ (quantum mechanics)

$|\Lambda| = L/a$ sites (discretized time). The Green's function $C(t,t) = 1/(2m_0)$ is **finite** (no UV divergence). The self-energy $\Sigma = O(1/N)$ (no log divergence).

Oscillations at the saddle width: $\sim L/(aN^2)$. For fixed L, a and large N: oscillations vanish. The saddle-point expansion converges for $N > C \cdot L/a$. The mass gap is perturbative.

4.3 $d = 2$ (QFT)

$|\Lambda| = (L/a)^2$ sites. The Green's function $C(x,x) = \frac{1}{4\pi} \log \frac{\Lambda^2}{m_0^2}$ **diverges** logarithmically with the UV cutoff.

The self-energy: $\Sigma \sim \frac{1}{N} \cdot \log \frac{\Lambda}{m_0} \sim \frac{1}{N} \cdot N = O(1)$. The $1/N$ from BL is compensated by the log divergence — **dimensional transmutation**. The gap equation gives $m_0 \sim \Lambda\, e^{-c/g^2}$ (exponentially small).

Oscillations at the saddle width: $\sim |\Lambda|/N^2 = L^2/(a^2 N^2)$. For the continuum limit $a \to 0$: oscillations grow, sign problem becomes severe.

4.4 Summary

	$d = 0$	$d = 1$	$d = 2$
$C(x, x)$	$1/m_0^2$	$1/(2m_0)$	$\frac{1}{4\pi} \log \frac{\Lambda}{m_0}$
UV divergence	none	none	**log**
Self-energy Σ	$O(1/N)$	$O(1/N)$	$\mathbf{O(1)}$
Gap equation	trivial	perturbative	**dim. transmutation**
Oscillation at width	$O(1/N^2)$	$O(L/(aN^2))$	$O(L^2/(a^2N^2))$
Sign problem	none	mild	**severe**

The fundamental difference: in $d = 2$, the log divergence of $C(x, x)$ makes the self-energy $O(1)$ instead of $O(1/N)$, producing dimensional transmutation and making the sign problem unavoidable.

5 The saddle-point structure

Despite the difference in operator structure, both models share the same saddle-point analysis.

5.1 The gap equation

On the real σ-axis, the saddle of the exponent in (3) is at $\sigma = 0$ (after absorbing the saddle value into m_0^2 for the NLSM, or into the gap equation for the LSM). The gap equation in both cases:

$$(-\Delta + m_0^2)^{-1}_{xx} = \beta_{\text{eff}} \tag{10}$$

where β_{eff} encodes the coupling. In 2d: $m_0 \sim \Lambda\, e^{-c/g^2}$ (exponentially small).

5.2 The saddle-point Gaussian

Near $\sigma = 0$: the exponent is approximately $-\frac{1}{2}(\sigma, H\sigma)$ where $H > 0$ (positive definite Hessian, including both the bare $1/(2\lambda)$ and the $\operatorname{Tr}\log$ contribution). The saddle-point Gaussian IS log-concave on the real axis.

5.3 The oscillatory corrections

Away from $\sigma = 0$: the exponent develops an imaginary part $\operatorname{Im} f(\sigma) \neq 0$, causing oscillations. The sign problem is the difficulty of controlling these oscillations.

Key scales (at the saddle):

- Width of the Gaussian: $\sigma \sim 1/\sqrt{N}$ per site.
- Oscillation onset: $\operatorname{Im} f \sim O(\sigma^3/\sqrt{N})$ (cubic, since $\operatorname{Im} f'' = 0$ at the saddle).
- Total oscillation at the saddle width: $\sim |\Lambda|/N^2$.

For $N \gg \sqrt{|\Lambda|}$: oscillations are negligible within the saddle-point width. The saddle-point approximation is accurate.

6 What Kupiainen proved

6.1 NLSM (1980a): mass gap

Kupiainen [9] proved the mass gap for the lattice NLSM at large N: exponential decay of correlations in the odd sector, with mass $m(\beta, n) \to m_0(\beta)$ as $n \to \infty$.

The proof uses the constraint trick (m_0^2 added for free), the dual representation with complex measure, reflection positivity, chessboard estimates, and a random walk expansion. The threshold is $n > a_1 m_0^{-a_2}$ with $a_2 = 6(d+2)+1 = 25$ in $d = 2$.

6.2 LSM (1980b): pressure only

Kupiainen [10] studied the continuum $(\varphi^2)_2^2$ model (the LSM) and proved a **weaker** result: the $1/n$ expansion for the **pressure** (vacuum energy) is asymptotic:

$$p = \sum_{j=0}^{m-1} p_j n^{-j} + R_m, \qquad |R_m| \leq (3m)!\, \lambda^4 \alpha^m \Big(\frac{\lambda^\beta}{n}\Big)^m. \tag{11}$$

The effective small parameter is λ^β/n (with β close to 10), not $e^{c\lambda}/n$.

He did NOT prove the mass gap for the LSM. The paper works in the φ-representation (integration by parts, no HS) and in the dual representation (complex nonlocal measure, same sign problem). He mentions that the mass gap would need "a new cluster expansion which explicitly cancels the diverging pressure factors" — left as future work that was apparently never published.

6.3 Implications

- The **NLSM mass gap** at large N is proved (Kupiainen 1980a).
- The **LSM mass gap** at large N is **open** — only the pressure expansion is known (Kupiainen 1980b).
- Proving the LSM mass gap would be a **genuinely new result**, not a reformalization of existing work.
- The LSM and NLSM share the same IR physics, so the mass gap "should" transfer. But making this rigorous is nontrivial.

7 Strategy: Lefschetz thimble

The right contour deformation is not a rotation to the imaginary axis (which diverges), but the **Lefschetz thimble**: the steepest descent manifold through the dominant saddle point. This connects our problem to a well-developed mathematical framework (Picard-Lefschetz theory) and a large recent literature on sign problems in lattice QFT. See `sign-problem.tex` for a general survey.

7.1 Lefschetz thimbles: definition

The exponent $f : \mathbb{C}^{|\Lambda|} \to \mathbb{C}$ from (3) is holomorphic (away from the singularities of the $\log\det$). For each critical point σ_* of f, the **Lefschetz thimble** $\mathcal{J}_{\sigma_*}$ is the stable manifold of σ_* under the anti-holomorphic gradient flow:

$$\frac{d\sigma}{dt} = \overline{\nabla f(\sigma)}, \qquad \mathcal{J}_{\sigma_*} = \left\{\sigma(0) : \lim_{t\to+\infty} \sigma(t) = \sigma_*\right\}. \tag{12}$$

Key properties:

1. $\mathcal{J}_{\sigma_*}$ is an $|\Lambda|$-dimensional real submanifold of $\mathbb{C}^{|\Lambda|}$.
2. $\operatorname{Im} f = \operatorname{Im} f(\sigma_*)$ is **constant** on $\mathcal{J}_{\sigma_*}$ (the flow preserves the imaginary part).
3. $\operatorname{Re} f$ is **maximized** at σ_* on $\mathcal{J}_{\sigma_*}$ (steepest descent from the saddle).
4. The integrand e^f restricted to $\mathcal{J}_{\sigma_*}$ is proportional to a **positive** function (constant phase $e^{i\operatorname{Im} f(\sigma_*)}$, positive amplitude $e^{\operatorname{Re} f}$).

Our "steepest descent contour" $\Gamma = \{\mathrm{Im}f = 0,\ \text{connected to } \sigma_* = 0\}$ is precisely the Lefschetz thimble $\mathcal{J}_0$ for the dominant saddle at $\sigma_* = 0$ (where $f(0)$ is real, so $\mathrm{Im}f(\sigma_*) = 0$).

7.2 Thimble decomposition

By Picard-Lefschetz theory, the original integration cycle decomposes as:

$$[\mathbb{R}^{|\Lambda|}] = \sum_{\sigma_*} n_{\sigma_*}\, [\mathcal{J}_{\sigma_*}], \tag{13}$$

where $n_{\sigma_*} \in \mathbb{Z}$ are intersection numbers (of $\mathbb{R}^{|\Lambda|}$ with the dual ascending manifolds $\mathcal{K}_{\sigma_*}$). Therefore:

$$Z = \sum_{\sigma_*} n_{\sigma_*}\, e^{i\operatorname{Im} f(\sigma_*)}\, Z_{\sigma_*}, \qquad Z_{\sigma_*} = \int_{\mathcal{J}_{\sigma_*}} e^{\mathrm{Re}f(\sigma) - \mathrm{Re}f(\sigma_*)}\, |d\sigma| > 0. \tag{14}$$

Each Z_{σ_*} is **real and positive**: the sign problem is solved per-thimble. But different thimbles contribute different phases $e^{i\operatorname{Im} f(\sigma_*)}$, creating a residual *inter-thimble sign problem* if multiple thimbles contribute.

7.3 Single-thimble dominance at large N

The exponent scales as $f \sim N \cdot g(\sigma)$ for a function g independent of N (since f contains the $N/2$ prefactor of $\operatorname{Tr}\log$ and the $\sigma^2/(4\lambda)$ term scales with the gap equation). Therefore:

- The Hessian scales as $f'' \sim N$: saddle-point width $\sim 1/\sqrt{N}$.
- At competing saddle points σ_*': $\mathrm{Re}f(\sigma_*) - \mathrm{Re}f(\sigma_*') \sim N \cdot \Delta g > 0$ (the action gap grows linearly in N).

- The dominant thimble contribution exceeds all others by e^{cN} for some $c > 0$.

Conclusion: For N sufficiently large, a single thimble captures Z up to exponentially small corrections:

$$Z = n_0\, Z_0 \left(1 + O(e^{-cN})\right). \tag{15}$$

This eliminates both the intra-thimble sign problem (the integrand is positive on $\mathcal{J}_0$) and the inter-thimble sign problem (other thimbles are exponentially suppressed).

7.4 No topological obstruction for scalar fields

Lawrence-Yamauchi [13] proved that for 2D Abelian lattice gauge theory, the sign problem *cannot* be removed by any contour deformation. The obstruction is topological: $\pi_1(U(1)) = \mathbb{Z}$ creates winding sectors with irreconcilable phases.

This obstruction does **not** apply to scalar field theories: the field space $\mathbb{R}^n$ is contractible ($\pi_k(\mathbb{R}^n) = 0$ for all k), so there is no topological obstruction to deforming the contour. For the O(N) model, the question of whether contour deformation solves the sign problem is **analytic** (singularity avoidance, decay at infinity), not topological.

8 The constant shift and thimble geometry

The thimble has a remarkably simple leading-order form: a constant imaginary shift $v = v_*$. This section computes the thimble explicitly, first in $d = 0$, then at general $|\Lambda|$, and establishes its key geometric property (Lagrangian submanifold).

8.1 The $d = 0$ computation

Consider one site ($|\Lambda| = 1$) with N components of φ. After HS and integrating out φ:

$$f(\sigma) = -\frac{\sigma^2}{4\lambda} + i\sigma\rho^2 - \frac{N}{2}\log(i\sigma) + \text{const.} \tag{16}$$

The saddle equation $f'(\sigma) = 0$ gives $\sigma^2 - 2i\lambda\rho^2\sigma + \lambda N = 0$, with solutions:

$$\sigma_* = i\left[\lambda\rho^2 \mp \sqrt{\lambda^2\rho^4 + \lambda N}\,\right]. \tag{17}$$

Both saddle points are **purely imaginary**: $\sigma_* = iv_*$ with $v_* \in \mathbb{R}$.

Writing $\sigma = u + iv$ (u = real part, v = imaginary part), $f(\sigma_*)$ is real (since σ_* is purely imaginary), so $\mathrm{Im} f(\sigma_*) = 0$. The thimble equation $\mathrm{Im} f = 0$ becomes:

$$u\left[-\frac{v}{2\lambda} + \rho^2\right] = \frac{N}{2}\arctan\!\left(\frac{-u}{v}\right). \tag{18}$$

Expanding for small u: $\arctan(-u/v) \approx -u/v$, so dividing by u:

$$-\frac{v}{2\lambda} + \rho^2 = -\frac{N}{2v} \quad \Longrightarrow \quad v^2 - 2\lambda\rho^2 v - \lambda N = 0, \tag{19}$$

which gives $v = v_*$ (the saddle value). Near the saddle, **the thimble is a horizontal line** $\sigma = u + iv_*$ — a constant imaginary shift.

8.2 The operator on the shifted contour

On the constant-shift contour $\sigma = u + iv_*$ (i.e., $v = v_*$ for all sites), with v_* chosen via the gap equation so that $-2v_*z = m_0^2$, the φ-operator becomes:

$$-\Delta + 2i\sigma z = -\Delta + 2i(u + iv_*)z = -\Delta + m_0^2 + 2iuz. \tag{20}$$

(We write $\alpha \equiv v_*$ when convenient.) The constant shift introduces the **real mass** m_0^2 into the operator. For each real field $u(x)$, the operator $-\Delta + m_0^2 + 2iuz$ has positive real part $-\Delta + m_0^2$, and the FK bound gives:

$$\left|(-\Delta + m_0^2 + 2iuz)^{-1}(x, 0)\right| \leq (-\Delta + m_0^2)^{-1}(x, 0) \sim e^{-m_0|x|}. \tag{21}$$

This is exactly Kupiainen's NLSM bound, achieved by contour deformation rather than by the constraint.

8.3 The thimble unifies LSM and NLSM

The constant imaginary shift has a conceptual interpretation:

- For the **NLSM**: the constraint $|\varphi|^2 = N\beta$ allows adding m_0^2 for free. No contour deformation needed.
- For the **LSM**: the add-subtract trick cancels m_0^2 on the real σ-axis ($v = 0$). But the *thimble* passes through $\sigma_* = iv_*$, and on the thimble, m_0^2 reappears as the real part $-2v_*z$.
- The contour deformation $v = 0 \to v = v_*$ **does for the LSM what the constraint does for the NLSM**: it introduces a real mass into the φ-operator.

8.4 The $N = \infty$ thimble is exactly flat

At the Gaussian (quadratic) level, the Hessian at the saddle is $H = -f''(\sigma_*)$, which is **real and positive definite** (since the operator at the saddle is $-\Delta + m_0^2$, real). Writing $\sigma = u + iv$ and expanding around $\sigma_* = iv_*$ (so $\delta u = u$, $\delta v = v - v_*$):

$$\mathrm{Im} f = -(\delta u,\, H\, \delta v) \quad \text{(at quadratic order)}. \tag{22}$$

Setting $\mathrm{Im} f = 0$ with $H > 0$: this forces $\delta v = 0$, i.e., $v = v_*$ everywhere. The $N = \infty$ thimble is the hyperplane $\sigma(x) = u(x) + iv_*$, **exactly flat**.

Since the Gaussian approximation is exact at $N = \infty$, the thimble is exactly the constant shift at leading order. Corrections come from the cubic and higher terms in f, suppressed by $1/\sqrt{N}$.

8.5 The leading $1/N$ correction: explicit formula

The cubic term of f at the saddle determines the first correction. Expanding $\operatorname{Tr}\log(A_0 + 2i\delta\sigma z)$ in powers of $\delta\sigma$:

$$f'''_{xyz} = \frac{8i}{\sqrt{N}} G_0(x-y)\, G_0(y-z)\, G_0(z-x), \tag{23}$$

where $G_0(k) = (k^2 + m_0^2)^{-1}$ is the massive propagator at the saddle. The cubic coefficient is **purely imaginary** (the factor i comes from $(2iz)^3 = -8i/N^{3/2}$, and G_0 is real at the saddle). Writing $f''' = iR$ with $R_{xyz} = (8/\sqrt{N})\, G_0(x{-}y)\, G_0(y{-}z)\, G_0(z{-}x)$ (the **triangle diagram**).

The thimble $v = v(u)$ is parameterized as $v_i = \frac{1}{2}\sum_{jk} S_{ijk}\, u_j u_k$ (quadratic in u, S symmetric in j, k). The stable manifold invariance condition $D\Phi(u)\cdot\dot{u} = \dot{v}|_{v=\Phi(u)}$ (with $\dot{u} = -Hu$ at leading order, $\dot{v} = Hv - \frac{1}{2}R[u, u, \cdot]$) gives a linear equation for S:

$$S(Hw_1, w_2) + S(w_1, Hw_2) + H \cdot S(w_1, w_2) = R(w_1, w_2) \qquad \text{for all } w_1, w_2 \in \mathbb{R}^{|\Lambda|}, \tag{24}$$

where the first two terms have H acting on an *input* of S, and the third has H acting on the *output*.

In Fourier space — with p, q the lattice momenta of the two u-inputs and $p{+}q$ the momentum of the v-output — equation (24) becomes algebraic: $\hat{S}(p, q)\, [\hat{H}(p) + \hat{H}(q) + \hat{H}(p{+}q)] = \hat{R}(p, q)$, giving:

$$\boxed{\hat{S}(p, q) = \frac{\hat{R}(p, q)}{\hat{H}(p) + \hat{H}(q) + \hat{H}(p+q)}} \tag{25}$$

The denominator is the sum of three σ inverse propagators at the three legs of the triangle vertex. The total symmetry in $p, q, -(p{+}q)$ confirms the Lagrangian property (Section 8.8). The ingredients:

$$\hat{H}(k) = \frac{1}{2\lambda} + 2B(k), \qquad B(k) = \frac{1}{|\Lambda|}\sum_l G_0(l)\, G_0(k{-}l) \quad \text{(bubble)}, \tag{26}$$

$$\hat{R}(p, q) = \frac{8}{\sqrt{N}} T(p, q), \qquad T(p, q) = \frac{1}{|\Lambda|}\sum_l G_0(l)\, G_0(p{-}l)\, G_0(q{+}l) \quad \text{(triangle)}. \tag{27}$$

The explicit thimble correction in Fourier space:

$$\hat{v}(k) = \frac{4}{\sqrt{N}\,|\Lambda|} \sum_{p+q=k} \frac{T(p, q)}{\hat{H}(p) + \hat{H}(q) + \hat{H}(p{+}q)}\, \hat{u}(p)\, \hat{u}(q). \tag{28}$$

8.6 Properties of the correction

1. **Standard Feynman diagram**: The kernel $T/(\hat{H} + \hat{H} + \hat{H})$ is the triangle vertex dressed by three σ-propagators $\hat{H}^{-1}$. This is a standard one-loop diagram in the $1/N$ expansion — the thimble shape *is* the diagrammatic expansion.

2. **No singularities**: The denominator $\hat{H}(p) + \hat{H}(q) + \hat{H}(p+q)$ is strictly positive (each $\hat{H}(k) > 0$), so $\hat{S}$ is smooth.

3. **Locality**: In position space, the kernel $K(x-y, x-z) \sim e^{-\min(m_0, M_\sigma)(|x-y|+|x-z|)}$ decays exponentially. The thimble correction at site x depends mainly on $u(y)$ for $|y-x| \lesssim 1/m_0$.

4. **Ultra-local limit**: For zero momentum ($p = q = k = 0$): $\hat{S}(0,0) = \hat{R}(0,0)/(3\hat{H}(0))$, recovering the $d = 0$ result $v = u^2/(6\hat{H}(0)\sqrt{N})$.

5. **Size at the saddle width**: The fluctuation $u \sim 1/\sqrt{N}$, so:

Quantity	Scale
$u(x)$ (real fluctuation)	$1/\sqrt{N}$
$v(x)$ (thimble correction)	$1/N^{3/2}$
v/u ratio	$1/N$
Residual Imf on constant shift	$u^3/\sqrt{N} \sim 1/N^2$ per site

The thimble is very nearly flat within the Gaussian peak.

8.7 Multi-site: constant shift plus local corrections

For the multi-site case ($|\Lambda| > 1$), the full thimble $\sigma = u + iv$ has:

$$v(x) = v_* + \frac{4}{\sqrt{N}} \sum_{y,z} K(x-y,\, x-z)\, u(y)\, u(z) + O(1/N), \tag{29}$$

where K is the position-space kernel of $T/(\hat{H} + \hat{H} + \hat{H})$. The leading-order form is the **constant shift** $v = v_*$ (i.e., $\sigma = u + iv_*$); the correction is local, quadratic in u, and suppressed by $1/\sqrt{N}$.

8.8 The thimble is a Lagrangian submanifold

The thimble $\mathcal{J}_0$ is a **Lagrangian submanifold** of $(\mathbb{C}^{|\Lambda|}, \omega)$, where $\omega = \sum_j du_j \wedge dv_j$ is the standard symplectic form on $\mathbb{C}^{|\Lambda|} \cong \mathbb{R}^{2|\Lambda|}$.

8.8.1 Proof

The anti-holomorphic gradient flow (12), written in real coordinates $\sigma = u + iv$, is:

$$\frac{du}{dt} = \frac{\partial(\mathrm{Re}f)}{\partial u} = \frac{\partial(\mathrm{Im}f)}{\partial v}, \qquad \frac{dv}{dt} = \frac{\partial(\mathrm{Re}f)}{\partial v} = -\frac{\partial(\mathrm{Im}f)}{\partial u}, \tag{30}$$

where the second equalities are Cauchy-Riemann. This is the **Hamiltonian flow** of Imf with respect to ω. Hamiltonian flows preserve the symplectic form.

Let X, Y be tangent vectors to $\mathcal{J}_0$ at a point σ. Under the flow, they remain tangent to $\mathcal{J}_0$ (flow-invariance of the stable manifold). As $t \to +\infty$, both flow into the stable subspace at σ_*, which is $\{v = 0\}$ (the real hyperplane). Since $\omega|_{\{v=0\}} = 0$:

$$\omega(X, Y) = \omega(X(t), Y(t)) \xrightarrow{t\to\infty} 0. \tag{31}$$

But $\omega(X(t), Y(t))$ is constant (Hamiltonian flow preserves ω), so $\omega(X, Y) = 0$. □

8.8.2 Consequences

1. **Generating functional**: Since $\mathcal{J}_0$ is Lagrangian, the thimble correction satisfies $v = \nabla_u \Phi$ for a single scalar functional $\Phi : \mathbb{R}^{|\Lambda|} \to \mathbb{R}$. The entire thimble is determined by Φ.

2. **Jacobian phase**: The induced volume form on a Lagrangian graph $\{(u, \nabla\Phi(u))\}$ has Jacobian $\det(I + i\nabla^2\Phi)$. Since $\nabla^2\Phi$ is symmetric with real eigenvalues λ_j:

$$|\det J| = \prod_j \sqrt{1 + \lambda_j^2}, \qquad \arg(\det J) = \sum_j \arctan(\lambda_j) = \operatorname{Tr}\arctan(\nabla^2\Phi). \tag{32}$$

   The Jacobian phase is **not zero** in general: a real symmetric matrix does not have $\pm$-paired eigenvalues. For $\nabla^2\Phi \sim u/\sqrt{N}$ (from the leading thimble correction): $\arg(\det J) \approx \operatorname{Tr}(\nabla^2\Phi) \sim |\Lambda|/N$. See Section 9.5.3 for how to handle this.

3. **Integrability of the $1/N$ correction**: The total symmetry of $\hat{S}(p, q)$ in all three momenta $p, q, -(p+q)$ (from (25)) is precisely the condition $\partial v_i/\partial u_j = \partial v_j/\partial u_i$, i.e., the existence of Φ.

9 The Hamilton-Jacobi equation

The Lagrangian property reduces the thimble to a single scalar functional $\Phi(u)$. This section derives the exact equation for Φ, its perturbative expansion, and the convergence theory.

9.1 The equation

The condition $\operatorname{Im} f = 0$ on the thimble $v = \nabla\Phi$ gives a **single scalar equation** for a single scalar functional:

$$\boxed{\operatorname{Im} f\big(u + i\nabla_u\Phi(u)\big) = 0 \qquad \text{for all } u \in \mathbb{R}^{|\Lambda|}.} \tag{33}$$

This is a Hamilton-Jacobi equation: u is the "position," $\nabla\Phi$ is the "momentum," $\operatorname{Im} f$ is the "Hamiltonian," and Φ is the "action" (generating function of the Lagrangian submanifold).

9.2 Concrete form: the arctan equation

Writing $\psi(x) = \partial\Phi/\partial u(x)$ for the thimble correction field, the imaginary part is $v(x) = v_* + \psi(x)$ and $\sigma = u + iv$. The φ-operator on the thimble is:

$$-\Delta + 2i\sigma z = \underbrace{\left(-\Delta + m_0^2 - \tfrac{2\psi}{\sqrt{N}}\right)}_{M(\psi)} + i\,\underbrace{\tfrac{2u}{\sqrt{N}}}_{V(u)}, \tag{34}$$

where $M(\psi)$ is a real Schrödinger operator (with potential $m_0^2 - 2\psi/\sqrt{N}$) and $V(u)$ is a real multiplication operator. Assuming $M > 0$ (the thimble stays above the spectral gap):

$$\operatorname{Im}\operatorname{Tr}\log(M + iV) = \operatorname{Tr}\arctan\bigl(M^{-1/2}\,V\,M^{-1/2}\bigr), \tag{35}$$

where arctan acts on the operator $D = M^{-1/2}VM^{-1/2}$ (self-adjoint, real eigenvalues d_k) by $\arctan(D) = \sum_k \arctan(d_k)\,|e_k\rangle\langle e_k|$.

The exact HJ equation (33) becomes:

$$\boxed{\frac{1}{2\lambda}\sum_x u(x)\bigl(v_*{+}\psi(x)\bigr) + \frac{N}{2}\operatorname{Tr}\arctan\Bigl(M(\psi)^{-1/2}\,\frac{2u}{\sqrt{N}}\,M(\psi)^{-1/2}\Bigr) + \rho^2\sum_x u(x) = 0} \tag{36}$$

for all $u \in \mathbb{R}^{|\Lambda|}$, with $\psi = \nabla\Phi(u)$ and $M(\psi) = -\Delta + m_0^2 - 2\psi/\sqrt{N}$.

Remark 1 (Structure of the equation)*.* Equation (36) is **exact at finite** N — no expansion in $1/N$. The three terms have clear roles:

- $\frac{1}{2\lambda}(u, v_*{+}\psi)$: the bare Gaussian contribution to $\operatorname{Im} f$.
- $\frac{N}{2}\operatorname{Tr}\arctan(\cdots)$: the determinantal contribution from integrating out φ. The arctan is the **phase** of $\det(M + iV)^{1/2}$ — the argument of the complex determinant.
- $\rho^2 \sum u$: the linear coupling to the HS source.

At $\psi = 0$ (constant shift) and $u = 0$: the equation reduces to the **gap equation** for m_0 (the arctan linearizes to $M_0^{-1}V$ and the trace gives $G_0(0)\sum u$).

9.3 Perturbative expansion

Expanding $\Phi = \Phi_3 + \Phi_4 + \cdots$ in powers of u (equivalently, $1/\sqrt{N}$):

$$\Phi_3(u) = \frac{1}{6}\sum_{ijk} S_{ijk}\,u_iu_ju_k \qquad \text{(triangle diagram, Section 8.5)}, \tag{37}$$

$$\Phi_4(u) = \frac{1}{24}\sum_{ijkl} S^{(4)}_{ijkl}\,u_iu_ju_ku_l \qquad \text{(box diagram)}, \tag{38}$$

$$\vdots$$

There is no Φ_1 or Φ_2: $\Phi_1 = 0$ by the gap equation (the linear term in (36) vanishes at $u = 0$), and $\Phi_2 = 0$ because the Hessian $f''(\sigma_*)$ is real (the quadratic term in $\operatorname{Im} f$ vanishes identically). Each Φ_n for $n \geq 3$ is determined by matching order u^n in (36), giving $S^{(n)}$ in terms of n-loop Feynman diagrams and lower-order data.

9.4 Convergence of the $1/N$ expansion

Proposition 2 (Analyticity of the thimble)**.** *For $|\Lambda| < \infty$ and $m_0 > 0$, the generating functional $\Phi(u)$ is real-analytic in $u \in \mathbb{R}^{|\Lambda|}$. The Taylor series $\Phi = \sum_{n\geq 3} \Phi_n$ converges in the **sup-norm** $\|u\|_\infty < r$ with radius:*

$$r \;\geq\; c\, m_0\, N^{1/4}, \tag{39}$$

*where $c > 0$ depends on λ but **not** on $|\Lambda|$ or N.*

Proof sketch. The thimble $\mathcal{J}_0$ is the stable manifold of σ_* under the anti-holomorphic gradient flow (12). Since f is holomorphic and the saddle is non-degenerate ($H > 0$), the **analytic stable manifold theorem** guarantees that Φ is real-analytic in u.

The radius is limited by the nearest singularity of Φ, which occurs when $M(\psi) = -\Delta + m_0^2 - 2\psi/\sqrt{N}$ hits zero eigenvalue. This requires $|\psi(x)| \geq m_0^2\sqrt{N}/2$ *at some site x* — a **pointwise** (sup-norm) condition, independent of $|\Lambda|$. Since $p \sim u^2$, the singularity is at $\|u\|_\infty \sim m_0 N^{1/4}$. □

9.5 The Jacobian phase and the order of limits

9.5.1 The residual phase problem

On any contour $\sigma = u + i\nabla\Phi(u)$, the integral acquires a Jacobian:

$$\int_{\mathcal{J}_0} e^f\, d\sigma = \int_{\mathbb{R}^{|\Lambda|}} e^{f(u+i\nabla\Phi)}\, \det(I + i\nabla^2\Phi)\, du. \tag{40}$$

The total phase of the integrand is:

$$\text{total phase} = \underbrace{\mathrm{Im} f(u + i\nabla\Phi)}_{\text{action phase}} \;+\; \underbrace{\mathrm{Tr}\arctan(\nabla^2\Phi)}_{\text{Jacobian phase}}. \tag{41}$$

The classical HJ equation (36) sets the action phase to zero. But the Jacobian phase $\mathrm{Tr}\arctan(\nabla^2\Phi) \neq 0$: a real symmetric matrix does *not* have $\pm$-paired eigenvalues, so $\arg\det J \neq 0$.

For the leading correction $\nabla^2\Phi \sim u/\sqrt{N}$: $\mathrm{Tr}\arctan(\nabla^2\Phi) \approx \mathrm{Tr}(\nabla^2\Phi) \sim |\Lambda|/N$, with variance $\sim |\Lambda|/N^2$.

Contour	Action phase	Jacobian phase
Constant shift ($v = v_*$)	$O(\lvert\Lambda\rvert/N^2)$	0 (flat)
Classical thimble ($\mathrm{Im} f = 0$)	0	$O(\lvert\Lambda\rvert/N^2)$

Both have **the same total phase** $O(|\Lambda|/N^2)$. The classical thimble moves the phase from the action to the Jacobian but does not eliminate it.

9.5.2 Consequence for the mass gap

The triangle inequality gives: $|Z^{-1}\int G_\sigma e^f\, d\sigma| \leq G_{m_0}(x, 0) \cdot (\text{average sign})^{-1}$, where the average sign $\sim e^{-c|\Lambda|/N^2}$. For $|\Lambda| \to \infty$ at fixed N, this bound diverges. The contour shift (or classical thimble) alone requires $N \gg \sqrt{|\Lambda|}$.

9.5.3 The quantum Hamilton-Jacobi equation

To eliminate *both* phases, we must solve the **quantum HJ equation**:

$$\boxed{\operatorname{Im} f\big(u + i\nabla\Phi\big) + \operatorname{Tr}\arctan(\nabla^2\Phi) = 0 \qquad \text{for all } u \in \mathbb{R}^{|\Lambda|}.} \tag{42}$$

This incorporates the Jacobian measure into the phase condition. The integrand in (40) then has **exactly zero total phase**: $e^f \det J$ is real and positive on the quantum thimble.

Equation (42) is a **second-order** PDE in Φ (due to the $\nabla^2\Phi$ in the $\operatorname{Tr}\arctan$), in contrast to the first-order classical HJ equation (36). Existence of solutions is more subtle: the classical thimble provides a natural starting point, with the quantum correction $\delta\Phi = O(1/N)$ absorbing the Jacobian phase.

The quantum HJ equation has the same perturbative structure: $\Phi = \Phi_3 + \Phi_4 + \cdots$ with each order modified by $O(1/N)$ relative to the classical HJ solution.

9.5.4 Bounds-only approach to the quantum thimble

Just as for the classical HJ (Section 10), we do not need to *solve* the quantum HJ equation — we need **existence** of a solution Φ with **a priori bounds**. On the quantum thimble, the full integrand $e^f \det J$ is real and positive (both phases cancel by construction). Define the **effective potential**:

$$V_{\text{eff}}(u) = -\operatorname{Re} f\big(u + i\nabla\Phi(u)\big) - \log|\det(I + i\nabla^2\Phi(u))|. \tag{43}$$

The BL measure is $e^{-V_{\text{eff}}(u)}\,du$ — real, positive, no residual phase. The three a priori bounds needed are:

1. **Proximity**: $|\psi(x)| \le C_1\|u\|_\infty^2/\sqrt{N}$ (same as classical — the quantum correction is $O(1/N)$ smaller).

2. **Spectral gap**: $M(\psi) \ge m_0^2/2 > 0$ (same bound, same proof).

3. **Hessian of V_{eff}**: $\nabla^2 V_{\text{eff}} \ge \kappa N$. This differs from the classical case by the $\log|\det J|$ contribution, which modifies the Hessian by $O(1)$ (from $\nabla^2\operatorname{Tr}\log \sim \nabla^2\operatorname{Tr}(\nabla^2\Phi) \sim S'''$, the triangle diagram derivative). For N large, the κN term dominates.

The quantum Φ differs from the classical Φ by $O(1/N)$, so the bounds are the same to leading order. The key advantage: **zero total phase** means the average sign is exactly 1, and the mass gap bound is **uniform in** $|\Lambda|$ without any $N \gg \sqrt{|\Lambda|}$ condition.

Proposition 3 (Existence of the quantum thimble)**.** *Define $F(\Phi)(u) = \operatorname{Im} f(u + i\nabla\Phi) + \operatorname{Tr}\arctan(\nabla^2\Phi)$. For $N > N_0(m_0, \lambda)$ on a finite lattice Λ:*

1. *$F(0)(u) = O(\|u\|_\infty^3/\sqrt{N})$ per site (the residual phase at the constant shift).*

2. *The linearization $DF(0)$ is invertible: the leading term is $\partial(\operatorname{Im} f)/\partial\psi \cdot \nabla(\cdot) + \Delta_u(\cdot)$, which is controlled by the Hessian $H > 0$.*

3. *By the **implicit function theorem**: there exists Φ with $F(\Phi) = 0$ and $\|\nabla\Phi\|_\infty \le C\,\|u\|_\infty^2/\sqrt{N}$ (the proximity bound).*

Proof sketch. F is smooth (from the analyticity of f and arctan). $F(0)$ is explicitly computable and small: $\mathrm{Im} f(u + iv_*)$ is $O(u^3/\sqrt{N})$ per site (the cubic term from the $\mathrm{Tr}\log$), and $\mathrm{Tr}\arctan(0) = 0$. The Fréchet derivative $DF(0)$ maps $\delta\Phi \mapsto \partial(\mathrm{Im} f)/\partial v \cdot \nabla(\delta\Phi) + \mathrm{Tr}(\nabla^2(\delta\Phi))$. The first term involves the Hessian H (positive definite), so $DF(0)$ is a bounded invertible operator on a space of smooth functionals. By the quantitative IFT: $\|\Phi\| \le C\,\|F(0)\|/\|DF(0)^{-1}\| = O(u^3/\sqrt{N})$. The spectral gap and Hessian bounds follow from proximity to the classical solution (which satisfies them). □

Remark 4 (Lean formalization). For the initial formalization, we **axiomatize** Proposition 3 (existence of Φ with bounds). The IFT proof is a realistic future target: Mathlib has implicit function theorems, and the hypotheses (smoothness from `contDiffAt_log_det`, invertibility from $H > 0$) are available.

9.5.5 The transfer matrix resolution

A more robust approach avoids the spacetime phase problem entirely by reducing the mass gap to a **spectral gap** of the transfer matrix.

On a lattice $\Lambda = \Lambda_s \times \{0, \dots, T-1\}$ (spatial sites $\times$ time), the partition function factorizes via the transfer matrix $\mathcal{T}$:

$$Z = \mathrm{Tr}(\mathcal{T}^T), \qquad \langle \varphi(x,t)\varphi(0,0)\rangle = \frac{\mathrm{Tr}(\mathcal{T}^{T-t}\,\hat\varphi(x)\,\mathcal{T}^t\,\hat\varphi(0))}{\mathrm{Tr}(\mathcal{T}^T)}. \tag{44}$$

The mass gap is $m = \log(\mathrm{ev}_0/\mathrm{ev}_1)$, where $\mathrm{ev}_0 > \mathrm{ev}_1$ are the two largest eigenvalues of $\mathcal{T}$. This is a property of $\mathcal{T}$ alone — **independent of** T (the time extent).

The contour shift / thimble analysis is then applied to the **spatial** integral that defines $\mathcal{T}$, which involves only $|\Lambda_s|$ sites (the spatial volume). For a 2D theory: $|\Lambda_s| = L/a$ (one spatial dimension), and $N \gg \sqrt{|\Lambda_s|} = \sqrt{L/a}$ is a condition on N vs. the *spatial* volume, not the full spacetime volume.

Key advantage: The spectral gap of $\mathcal{T}$ is an intrinsic property of the one-step transfer matrix. The phase problem involves $|\Lambda_s|$ sites (fixed for a given spatial lattice), not $|\Lambda| = |\Lambda_s| \times T$ (which grows with the time extent). The mass gap is then:

1. Proved on a fixed spatial lattice Λ_s with $N \gg \sqrt{|\Lambda_s|}$ (contour shift for the spatial integral in $\mathcal{T}$).

2. Extended to $|\Lambda_s| \to \infty$ by showing the spectral gap is uniform (BL concentration + FK bound, both uniform in $|\Lambda_s|$ as argued in Sections 10–10.2.2).

3. This is exactly the structure of Kupiainen's proof: reflection positivity + transfer matrix + cluster expansion (replaced here by BL).

Remark 5 (Three routes to the thermodynamic limit). (a) **Transfer matrix + spatial contour shift**: phase problem involves $|\Lambda_s|$ only; $N \gg \sqrt{L/a}$. This is the recommended approach.

(b) **Quantum HJ equation**: eliminates phase exactly; requires second-order PDE existence theory.

(c) **Spacetime contour shift with $N \gg \sqrt{|\Lambda|}$**: simplest, but couples N to the full spacetime volume.

10 Solution theory and the mass gap proof

This section establishes the a priori bounds on Φ needed for the mass gap, addresses the technical issues (singularity avoidance, log-concavity, BL version), and assembles the proof.

10.1 Solution theory for the HJ equation

The HJ equation (36) is exact but highly nonlinear. We describe three approaches to its solution and the a priori bounds needed for the mass gap proof.

10.1.1 Method of characteristics

The HJ equation $H(u, \psi) = 0$ with "Hamiltonian" $H(u, \psi) = \mathrm{Im} f(u + i(v_* + \psi))$ has characteristic equations:

$$\frac{du}{ds} = \frac{\partial H}{\partial \psi}, \qquad \frac{d\psi}{ds} = -\frac{\partial H}{\partial u}. \tag{45}$$

Since the anti-holomorphic gradient flow (12) preserves $\mathrm{Im} f$, the characteristics (45) are precisely the **projections of the gradient flow** onto (u, ψ)-space. In other words:

Solving the HJ equation and solving the gradient flow are dual descriptions of the same object.

The gradient flow constructs the thimble as a stable manifold (dynamical systems perspective); the HJ equation constructs it as a Lagrangian graph (symplectic geometry perspective). The duality means that any estimate proved via the flow automatically gives an estimate on Φ, and vice versa.

10.1.2 A priori bounds on Φ

For the mass gap proof, we do not need the explicit form of Φ — we need three **a priori bounds**:

Proposition 6 (Thimble regularity)**.** *For $N > N_0$ and $\|u\|_\infty \leq m_0 N^{1/4}/C$ (inside the convergence disk):*

1. ***Proximity to constant shift**: $|\psi(x)| = \left|\dfrac{\partial \Phi}{\partial u(x)}\right| \leq \dfrac{C_1}{\sqrt{N}} \|u\|_\infty^2$.*

2. ***Spectral gap preservation**: $M(\psi) = -\Delta + m_0^2 - 2\psi/\sqrt{N} \ \geq\ m_0^2/2 > 0$.*

3. ***Hessian control***: $\|\nabla^2 \mathrm{Re}f|_{\mathcal{J}_0} - H\|_{\mathrm{op}} \le C_2/N$, *so the Hessian of* $\mathrm{Re}f$ *on the thimble is* $H + O(1/N)$.

Proof sketch. **Bound (1)** follows from the convergent Taylor expansion: $\psi = \nabla\Phi$ starts at order u^2 with coefficient $O(1/\sqrt{N})$ (from the triangle diagram), and higher orders are controlled by the convergence of the series.

Bound (2) follows from (1): the perturbation $2|\psi|/\sqrt{N} \le 2C_1\|u\|_\infty^2/N$, which is $\ll m_0^2$ within the saddle-point region $\|u\|_\infty \lesssim 1/\sqrt{N}$.

Bound (3): on the thimble $v = \nabla\Phi(u)$, the pullback of $\mathrm{Re}f$ to $\mathbb{R}^{|\Lambda|}$ (via $u \mapsto u + i\nabla\Phi$) has Hessian $H + O(1/N)$ corrections from the Φ-dependent terms in $\mathrm{Re}f$ (the cubic and higher terms contribute $O(1/N)$ to the second derivative at the saddle width). □

What these bounds give:

- Bound (1) + (2): the FK bound applies on the thimble. The operator $M(\psi)+iV(u) = -\Delta + m_0^2 + O(1/N) + iV$ has positive real part $\ge m_0^2/2$, giving $|G_\sigma(x,0)| \le C\, e^{-m_0|x|/\sqrt{2}}$ for every u-configuration on the thimble.
- Bound (3): Brascamp-Lieb applies on the thimble. The measure $e^{\mathrm{Re}f}\,du$ has Hessian $\ge \kappa N$ (with $\kappa > 0$ independent of $|\Lambda|$), giving $\mathrm{Var}(u(x)) \le 1/(\kappa N)$.

10.1.3 Fixed-point formulation

The HJ equation can be reformulated as a fixed-point problem. Define the map $\mathcal{T}$ by: given $\psi : \mathbb{R}^{|\Lambda|} \to \mathbb{R}^{|\Lambda|}$ (a candidate thimble correction), set

$$(\mathcal{T}\psi)(x) = -2\lambda\Big[\rho^2 + \frac{N}{2}\frac{\partial}{\partial u(x)}\mathrm{Tr}\arctan\big(M(\psi)^{-1/2}\,V(u)\,M(\psi)^{-1/2}\big)\Big] - v_*. \tag{46}$$

Then $\psi = \nabla\Phi$ solves the HJ equation if and only if ψ is a fixed point of $\mathcal{T}$.

For N sufficiently large: $\mathcal{T}$ is a contraction on the ball $\|\psi\|_\infty \le m_0^2\sqrt{N}/4$ in an appropriate function space (the arctan is Lipschitz with constant < 1 when the spectral gap of M is large). The Banach fixed-point theorem then gives:

1. **Existence**: a unique solution $\psi = \nabla\Phi$ in the ball.
2. **Constructive approximation**: the iterates $\psi^{(n+1)} = \mathcal{T}\psi^{(n)}$ (starting from $\psi^{(0)} = 0$, the constant shift) converge geometrically.
3. **Quantitative bounds**: $\|\psi - \psi^{(n)}\| \le C\,\theta^n$ with contraction rate $\theta < 1$.

This provides an alternative to the Taylor expansion: a **non-perturbative**, constructive proof that the thimble exists and satisfies the a priori bounds of Proposition 6.

10.1.4 What the proof needs vs. what we compute

For the mass gap proof, the HJ equation plays the role of an **existence and regularity theorem**, not a computational tool. Specifically:

Step	What is needed from the HJ equation
Positivity	$\mathrm{Im} f = 0$ on the thimble (by construction)
BL concentration	Hessian of $\mathrm{Re} f \geq \kappa N$ (Bound 3)
FK bound	$M(\psi) > 0$ on the thimble (Bound 2)
Single-thimble dominance	Analyticity + action gap $\sim N$

None of these require the explicit form of Φ — only bounds on Φ and its derivatives, all of which follow from the convergence of the expansion (or the fixed-point argument).

10.2 Properties of the integral over $\mathcal{J}_0$

On $\mathcal{J}_0$, the measure $e^{f(\sigma)}\,d\sigma$ (with the appropriate orientation) is:

1. **Real and positive** (since $\mathrm{Im}\, f = 0$ and $e^{\mathrm{Re}\, f} > 0$).
2. **Log-concave near the saddle**: $\mathrm{Re}\, f \approx -\frac{1}{2}(\sigma, H\sigma)$ (Gaussian, $H > 0$).
3. **Decaying**: $\mathrm{Re}\, f \to -\infty$ along $\mathcal{J}_0$ away from the saddle (steepest descent property).

We now address three technical issues in detail.

10.2.1 Singularity avoidance in the thermodynamic limit

The singularities of f (where $\det(-\Delta + 2i\sigma z) = 0$) occur at $\sigma = ik^2/(2z)$ — purely imaginary, at distance $\geq m_0^2\sqrt{N}/2$ from the saddle. On the thimble, Bound (2) of Proposition 6 gives $M(\psi) \geq m_0^2/2 > 0$, so the thimble *never reaches these singularities* within the convergence disk.

For extreme fluctuations: BL concentration gives

$$\Pr\big(\|u\|_\infty > t\big) \leq 2|\Lambda|\, e^{-\kappa N t^2/2}. \tag{47}$$

The singularity requires $|u| \sim m_0 N^{1/4}$ (the convergence radius). Substituting $t = m_0 N^{1/4}$:

$$\Pr\big(\text{thimble reaches singularity}\big) \leq 2|\Lambda|\, e^{-\kappa m_0^2 N^{3/2}/2}. \tag{48}$$

This is **superexponentially small in** N, uniformly in $|\Lambda|$ (since $e^{-cN^{3/2}}$ beats any polynomial in $|\Lambda|$ for N large). Singularity avoidance holds in the thermodynamic limit.

10.2.2 Global vs. local log-concavity

The Hessian of Ref on the thimble is $H + O(1/N)$ (Bound 3) *near the saddle*. Far from the saddle, the Tr log term can introduce non-convexity. The question: does BL require *global* log-concavity?

Answer: For our application, *local* log-concavity within the Gaussian peak suffices, for two reasons:

1. **The Gaussian dominates the tails**. The bare term $-\|u\|^2/(4\lambda)$ in Ref provides Gaussian decay. The Tr log correction to the Hessian is bounded by $\|(\partial^2/\partial u^2)\mathrm{Tr}\log M\| \leq C/m_0^4$ (from the resolvent bound $\|M^{-1}\| \leq 2/m_0^2$). The Gaussian Hessian $1/(2\lambda)$ dominates for $\lambda < m_0^4/(2C)$ — a condition on the coupling, not the volume.

2. **BL with a cutoff**. One can apply BL to the measure restricted to the log-concave region $\|u\|_\infty \leq R$ (with $R \sim m_0 N^{1/4}/C$, the convergence radius), and bound the contribution from $\|u\|_\infty > R$ by the Gaussian tail. Since the tail probability is $\leq 2|\Lambda|e^{-\kappa N R^2/2} \sim e^{-cN^{3/2}}$, the correction is negligible.

In fact, for λ small enough (weak coupling), Ref on the thimble IS globally strictly convex: the Gaussian $1/(2\lambda)$ term dominates the Tr log curvature everywhere. The threshold is $\lambda < c\, m_0^4$, which is satisfied in the weak-coupling regime relevant to the continuum limit.

10.2.3 Brascamp-Lieb: which version?

The measure $e^{\mathrm{Re} f}\, du$ on the thimble is a probability measure on $\mathbb{R}^{|\Lambda|}$ (after normalization). For BL, we need:

- **Standard BL** (Brascamp-Lieb 1976 [1]): applies to $e^{-V(u)}\, du$ with V strictly convex ($\nabla^2 V \geq \kappa > 0$). This works on a finite lattice if $V = -\mathrm{Re} f$ is globally strictly convex — which holds for $\lambda < c\, m_0^4$ (see above). **No extension needed** in this regime.

- **Conlon-Dabkowski extension** [2]: applies to lattice field models with uniformly convex *gradient* action (weaker than strict convexity of V). This would be needed if λ is not small, i.e., the non-convex Tr log tails are significant. Their result is stated for lattice models in $d \geq 2$ and gives near-Gaussian correlation bounds.

- **BL from markov-semigroups** (our Lean library): the `brascampLieb_poincare` theorem gives $\mathrm{Var}(f) \leq (1/(\kappa N))\,\mathbb{E}[\|\nabla f\|^2]$ for (κN)-log-concave measures. This is the version we would formalize. It requires `LogConcaveMeasure` with convexity parameter κN, which we construct from the Hessian bound $H \geq \kappa N$.

10.2.4 Uniformity and the infinite-volume limit

The mass gap is the spectral gap of the transfer matrix $\mathcal{T}$ (Section 9.5.5). The contour shift / BL analysis is applied to the **spatial** integral defining $\mathcal{T}$, which involves $|\Lambda_s|$ sites.

The proof requires $N \gg \sqrt{|\Lambda_s|}$ to control the residual phase. For a fixed spatial lattice (e.g., $|\Lambda_s| = L/a$ in 2d), this is a **fixed** condition $N > N_0(L/a, m_0, \lambda)$ independent of the time extent.

Taking $|\Lambda_s| \to \infty$ (spatial thermodynamic limit): The spectral gap of $\mathcal{T}$ must be shown uniform in $|\Lambda_s|$. This is a stronger statement requiring either:

(a) The quantum HJ equation (42) (eliminating the phase), or

(b) A direct proof that the spectral gap of $\mathcal{T}$ is controlled by the per-site BL + FK bounds (which are volume-independent), without passing through the phase.

Kupiainen's proof uses option (b) via a cluster expansion. The BL approach should give the same result: the FK bound is per-configuration (volume-independent), and BL concentration holds with κ independent of $|\Lambda_s|$ (the Hessian $\hat{H}(k) = 1/(2\lambda) + 2B(k)$ has a spectral gap converging to a finite limit as $|\Lambda_s| \to \infty$).

The remaining issue is the **phase control** in the spatial integral. For the transfer matrix, the relevant phase is $O(|\Lambda_s|/N^2)$, which vanishes for $N \gg \sqrt{|\Lambda_s|}$. In the continuum limit $a \to 0$ (so $|\Lambda_s| = L/a \to \infty$), this requires $N \gg \sqrt{L/a}$ — coupling N to the UV cutoff.

Remark 7 (The continuum limit is weak coupling). The global log-concavity condition $\lambda < c\, m_0^4$ might appear restrictive, since $m_0 \sim \Lambda\, e^{-c/g^2}$ is exponentially small. But the **continuum limit is automatically in the weak-coupling regime**: the renormalized coupling runs as $g^2(a) \sim 1/\log(\Lambda/m_0) \to 0$ as $a \to 0$. The bare coupling $\lambda \sim g^2/N$, while $m_0^4 \sim \Lambda^4\, e^{-4c/g^2}$, so:

$$\frac{\lambda}{m_0^4} \sim \frac{g^2}{N\Lambda^4\, e^{-4c/g^2}} \to 0 \qquad \text{as } a \to 0. \tag{49}$$

The condition $\lambda < c\, m_0^4$ is satisfied with **enormous margin** in the continuum limit. Global log-concavity is a consequence of asymptotic freedom, not an extra assumption.

The Conlon-Dabkowski extension [2] would be needed for strong coupling (fixed lattice spacing, λ not small) or for N close to the threshold N_0. Neither is relevant for the initial proof, which targets the continuum limit at large N.

10.3 The flowed manifold alternative

Instead of characterizing the exact thimble $\mathcal{J}_0$ (a global object determined by the full analytic structure of f), one can work with the **flowed manifold**:

$$\mathcal{M}_T = \left\{\sigma(T) : \sigma(0) \in \mathbb{R}^{|\Lambda|},\ \dot{\sigma} = \overline{\nabla f(\sigma)}\right\}, \tag{50}$$

obtained by flowing the real domain along (12) for time T. Properties:

- $\mathcal{M}_0 = \mathbb{R}^{|\Lambda|}$ (the original domain).
- $\mathcal{M}_T$ is diffeomorphic to $\mathbb{R}^{|\Lambda|}$ for all finite T (no topology change).
- $\mathcal{M}_T \to \bigcup_{\sigma_*} \mathcal{J}_{\sigma_*}$ as $T \to \infty$.
- The residual sign problem on $\mathcal{M}_T$ is bounded by $e^{-c(T)N}$ for a function $c(T)$ increasing in T.

Advantage: No need to identify critical points or compute intersection numbers. The flow (12) is a well-posed ODE system. For a rigorous proof, take T large enough that the residual oscillations are smaller than the mass gap signal.

This approach was developed computationally by Fukuma et al. [4, 5] ("Worldvolume HMC"), now extended to gauge theories [7] and the Hubbard model [6].

10.4 Proof plan (the quantum thimble argument)

The quantum thimble (Section 9.5.3) gives a remarkably simple proof of the mass gap:

1. **Contour deformation**: Deform the σ-integral from $\mathbb{R}^{|\Lambda|}$ to the quantum thimble, where the total integrand $e^f \cdot \det J$ is **real and positive** (quantum HJ eliminates both action and Jacobian phases). The two-point function becomes:

$$\langle\varphi(x)\varphi(0)\rangle = \frac{1}{Z}\int G_\sigma(x,0)\, e^{-V_{\text{eff}}(u)}\, du \tag{51}$$

   where $e^{-V_{\text{eff}}}$ is a positive measure on $\mathbb{R}^{|\Lambda|}$.

2. **FK bound (uniform in u)**: On the quantum thimble, the φ-operator is $-\Delta + m_0^2 + 2iuz$ with $u \in \mathbb{R}$. The FK/diamagnetic bound gives $|G_\sigma(x,0)| \le G_{m_0}(x,0)$ for **every** u-configuration. This is the operator monotonicity $|(\mathrm{M} + i\mathrm{V})^{-1}| \le \mathrm{M}^{-1}$.

3. **Trivial averaging**: Since $e^{-V_{\text{eff}}}$ is a **positive** measure and the FK bound is uniform in u:

$$|\langle\varphi(x)\varphi(0)\rangle| \le \frac{1}{Z}\int |G_\sigma|\, e^{-V_{\text{eff}}}\, du \le G_{m_0}(x,0)\cdot\frac{Z}{Z} = G_{m_0}(x,0). \tag{52}$$

   The ratio $Z/Z = 1$ because the measure is positive — **no sign problem, no average sign factor**.

4. **Green's function decay**: $G_{m_0}(x,0) = (-\Delta + m_0^2)^{-1}(x,0) \le (1/m_0^2)\, e^{-m_0|x|}$ (Fourier analysis on the torus).

5. **Mass gap**: Chaining steps 1–4: $|\langle\varphi(x)\varphi(0)\rangle| \le (1/m_0^2)\, e^{-m_0|x|}$ with $m_0 > 0$ from the gap equation, independent of $|\Lambda|$.

Remark 8 (The role of BL in the mass gap proof). The BL variance bound $\mathrm{Var}(u(x)) \le 1/(\kappa N)$ is not used in the mass gap chain (steps 2–4): the FK bound is uniform in u, so the average is controlled by the triangle inequality on the positive measure.

However, BL enters *indirectly*: the quantum thimble existence (step 1) requires the Hessian spectral gap $\kappa > 0$ via the implicit function theorem. The Hessian bound IS the BL convexity parameter. So the logical dependencies are:

$$\text{Hessian } \kappa > 0 \xrightarrow{\text{IFT}} \text{quantum thimble exists} \xrightarrow{\text{positive measure}} \text{FK bound averages trivially} \\ \to \text{mass gap.}$$

The Hessian bound $\hat{H}(k) = 1/(2\lambda) + 2B(k) \ge \kappa > 0$ is uniform in $|\Lambda|$ (the spectral gap converges to a finite limit). So the quantum thimble exists for all finite $|\Lambda|$ simultaneously, and the mass gap is uniform in volume.

Remark 9 (Volume independence). The constants m_0 and $1/m_0^2$ depend on the gap equation (coupling constants), **not on** $|\Lambda|$. The mass gap persists in the thermodynamic limit because:

- FK bound: per-configuration, no $|\Lambda|$ dependence.
- Positive measure: quantum HJ, ratio $Z/Z = 1$.
- Green's decay: m_0 from gap equation, independent of $|\Lambda|$.

For a rigorous proof, one can work with the *exact* constant shift (Cauchy's theorem for a flat contour is elementary) and treat the difference between the constant shift and the exact thimble as a perturbation. The residual $\mathrm{Im}\, f$ on the constant-shift contour is $O(u^3/\sqrt{N})$ per site — small within the saddle-point width but requiring control in the tails.

11 The diamagnetic inequality

The FK bound $|G_\sigma(x, 0)| \leq G_{m_0}(x, 0)$ used in Step 3 of the proof plan (§10.2) relies on the **diamagnetic inequality** for the resolvent of $M + iV$, where $M = -\Delta + m_0^2$ is positive definite and $V = 2uz$ is a real diagonal matrix. This section presents the semigroup proof, following Simon [14] Ch. 22 and Reed–Simon IV §X.4.

11.1 Statement

Theorem 10 (Diamagnetic inequality for finite matrices)**.** *Let M be a real symmetric positive definite $n \times n$ matrix with nonpositive off-diagonal entries (a Stieltjes matrix), and let V be a real diagonal matrix. Then for all x, y:*

$$|(M + iV)^{-1}(x, y)| \leq M^{-1}(x, y). \tag{53}$$

In particular, $M^{-1}(x, y) \geq 0$ for all x, y.

In our application, $M = -\Delta + m_0^2$ (the shifted lattice Laplacian) and $V = \mathrm{diag}(2u(x)/\sqrt{N})$, so the hypotheses are satisfied: M has off-diagonal entries -1 (from the nearest-neighbor Laplacian) and diagonal entries $2d + m_0^2 \geq m_0^2 > 0$, giving positive definiteness by Gershgorin.

11.2 The semigroup proof

The proof decomposes into five steps.

Step 1: Laplace transform representation. For any positive definite matrix A,

$$A^{-1} = \int_0^\infty e^{-tA}\, dt. \tag{54}$$

This follows from spectral decomposition: $A = \sum_k \lambda_k |e_k\rangle\langle e_k|$ with all $\lambda_k > 0$, so $\int_0^\infty e^{-t\lambda_k}\, dt = 1/\lambda_k$. Similarly, $(M + iV)^{-1} = \int_0^\infty e^{-t(M+iV)}\, dt$ since the eigenvalues of $M + iV$ have positive real part (from $M > 0$).

Step 2: Lie–Trotter product formula. For finite matrices A and B,

$$e^{A+B} = \lim_{n\to\infty} \big(e^{A/n}\, e^{B/n}\big)^n. \tag{55}$$

This is the Lie product formula; for matrices it follows from $e^{A/n}e^{B/n} = e^{(A+B)/n+O(1/n^2)}$ (Baker–Campbell–Hausdorff) and $(I + X/n)^n \to e^X$. Applied to $A = -tM$ and $B = -itV$:

$$e^{-t(M+iV)} = \lim_{n\to\infty} \big(e^{-tM/n}\, e^{-itV/n}\big)^n. \tag{56}$$

Step 3: Phase bound. Since $V = \mathrm{diag}(v_1, \ldots, v_n)$ is diagonal,

$$e^{-itV/n}(x, y) = \delta_{xy}\, e^{-itv_x/n} \tag{57}$$

has $|e^{-itV/n}(x, y)| = \delta_{xy}$. In particular, for any matrix P,

$$|(P\, e^{-itV/n})(x, y)| = |P(x, y)\, e^{-itv_y/n}| = |P(x, y)|. \tag{58}$$

Step 4: Heat kernel positivity. Since M has nonpositive off-diagonal entries (it is a *Z-matrix* or equivalently $-M$ is a *Metzler matrix*), the matrix exponential e^{-tM} has nonneg entries for all $t \geq 0$:

$$e^{-tM}(x, y) \geq 0 \quad \text{for all } x, y, t \geq 0. \tag{59}$$

Proof of (59). Write $-M = D+R$ where D is the diagonal part and $R \geq 0$ entrywise (since M has nonpositive off-diagonal). By the Euler approximation:

$$e^{-tM} = \lim_{n\to\infty} \Big(I - \frac{tM}{n}\Big)^n = \lim_{n\to\infty} \Big(I + \frac{t(D + R)}{n}\Big)^n. \tag{60}$$

For $n > t\|M\|$, each factor $I + t(D + R)/n$ has nonneg entries (diagonal entries $\geq 1 - t\|D\|_\infty/n > 0$, off-diagonal entries $= tR_{xy}/n \geq 0$). The product of entrywise-nonneg matrices is entrywise-nonneg, so $e^{-tM}(x, y) \geq 0$. □

Step 5: Assembly. Each Trotter factor satisfies:

$$|(e^{-tM/n}\, e^{-itV/n})(x, y)| \overset{(58)}{=} |e^{-tM/n}(x, y)| \overset{(59)}{=} e^{-tM/n}(x, y). \tag{61}$$

For the n-fold product, the entry-wise triangle inequality gives:

$$\begin{aligned} |\big(e^{-tM/n}e^{-itV/n}\big)^n(x, y)| &\leq \sum_{z_1,\ldots,z_{n-1}} \prod_{k=1}^{n} |(e^{-tM/n}e^{-itV/n})(z_{k-1}, z_k)| &(62)\\ &= \sum_{z_1,\ldots,z_{n-1}} \prod_{k=1}^{n} e^{-tM/n}(z_{k-1}, z_k) &(63)\\ &= (e^{-tM/n})^n(x, y) = e^{-tM}(x, y). &(64)\end{aligned}$$

Taking $n \to \infty$ (Trotter) and integrating over t (Laplace transform):

$$|(M+iV)^{-1}(x, y)| = \Big|\int_0^\infty e^{-t(M+iV)}(x, y)\, dt\Big| \leq \int_0^\infty |e^{-t(M+iV)}(x, y)|\, dt \leq \int_0^\infty e^{-tM}(x, y)\, dt = M^{-1}(x, y). \tag{65}$$

Remark 11 (M-matrix inverse nonnegativity). As a corollary, $M^{-1}(x, y) \geq 0$: from (54) and (59), $M^{-1} = \int_0^\infty e^{-tM}\, dt \geq 0$ entrywise. This is a standard result in M-matrix theory.

Remark 12 (The Lean formalization). The Lean file `DiagmagneticInequality.lean` (413 lines) decomposes this proof into 6 axioms corresponding to Steps 1–5 above, plus the M-matrix corollary. The phase bound (Step 3) and the entry-wise triangle inequality are fully proved; the remaining axioms (Trotter, heat kernel positivity, Laplace transform) use standard Mathlib tools (matrix exponential, spectral theorem).

12 Other strategies for comparison

Section 7 describes what we call **Strategy D** (Lefschetz thimble). For context, we compare with three alternatives.

12.1 Strategy A: NLSM (Kupiainen)

Work with the NLSM directly. The constraint gives real mass m_0^2 for free. The FK bound gives $|G| \leq G_{m_0}$ (single-σ decay). Cluster expansion + RP control the σ-average.

Advantage: Real mass in the operator, clean FK bound.
Disadvantage: Cluster expansion needed (technically heavy).
Threshold: $N > C\, m_0^{-a_2}$ with $a_2 = O(d^2)$ (Kupiainen Thm. 1).

12.2 Strategy B: LSM saddle-point (real axis)

Work with the LSM on the *real* σ-axis. The saddle-point Gaussian is log-concave (real, positive), but the full integrand has oscillatory corrections away from the saddle.

Idea: Apply BL to the saddle-point Gaussian and control the oscillatory corrections as a perturbation.

Challenge: The mass gap comes entirely from the σ-average (no single-σ decay). Need to show the oscillatory corrections don't destroy the saddle-point mass.

Advantage: Avoids cluster expansion (BL for the Gaussian).
Disadvantage: No FK bound; the sign problem is addressed by perturbative control, not eliminated.
Comparison with D: Strategy D *eliminates* the oscillations by deforming the contour; Strategy B tries to *control* them on the real axis.

12.3 Strategy C: LSM → NLSM equivalence

Prove that the LSM and NLSM have the same IR physics (they differ only at the σ-mass scale $M_\sigma \sim \sqrt{\lambda}\, v$), then use Strategy A.

Advantage: Gets the NLSM mass gap (with the real mass trick) from the LSM formulation.
Disadvantage: Proving IR equivalence rigorously is itself a major result.

12.4 Strategy comparison

	A (NLSM)	B (real axis)	C (equivalence)	D (thimble)
Sign problem	avoided	controlled	avoided	reduced/eliminated
Real mass m_0^2	from constraint	none	from constraint	from contour shift
FK bound	yes	no	yes	yes
BL needed	no	yes	no	yes
Cluster exp.	yes	no	yes	no
Proved	yes (K80)	no	no	no

Strategy D with the constant shift (Section 8) achieves the same operator structure as Strategy A ($-\Delta + m_0^2$ in the real part) without the constraint. The key insight: *the thimble does for the LSM what the constraint does for the NLSM.* The residual sign problem on the constant-shift contour is $O(|\Lambda|/N^2)$, controlled perturbatively for $N \gg \sqrt{|\Lambda|}$. This avoids both the cluster expansion (Strategy A) and the uncontrolled oscillations (Strategy B).

13 The $N = 2$ obstruction

For $N = 2$: the BKT transition occurs at finite λ (Dario-Garban [3]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). Any proof must exclude $N = 2$. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

14 Current status

14.1 Main theorem (in Lean)

41 files, 21 axioms, 1 sorry.

Theorem 13 (`ON_LSM_hasCorrelationDecay`)**.** *For the O(N) LSM interacting measure μ on a finite lattice with gap equation mass $m_0^2 > 0$:*

$$\mathtt{HasCorrelationDecay}\ \mu\ \mathrm{dist} \tag{66}$$

i.e., $\exists\, m, C > 0$ such that $|\langle \varphi^i(x)\varphi^i(y)\rangle_c| \leq C\, e^{-m\,\mathrm{dist}(x,y)}$ for all components i and sites x, y.

The proof chains two axioms:

1. `contour_deformation`: the connected correlator of the *specific* O(N) LSM measure `onInteractingMeasure` is bounded by $M^{-1}(x, y)$ where $M = -\Delta + m_0^2$.
2. `green_exponential_decay`: $M^{-1}(x, y) \leq (1/m_0^2)\, e^{-m_0\,\mathrm{dist}(x,y)}$.

14.2 What is proved (no axioms; 1 sorry in diagonal case)

- **HS identity**: from Mathlib's `fourierIntegral_gaussian`.
- **HS equivalence** (one-site): $\int d\sigma\, e^{-S_{\mathrm{HS}}} = c \cdot e^{-S_{\mathrm{original}}}$ — **fully proved** (was sorry, now via `push_cast; ring`).

- **Gap equation algebra**: $-2v_* z = m_0^2$, $v_* < 0$, singularity distance $|v_*| = m_0^2\sqrt{N}/2$.
- **Shifted operator**: $M = -\Delta + m_0^2$ has spectral gap $\geq m_0^2$ (from `psd_add_scalar_bound`).
- **Phase cancellation**: $r_1 e^{i\theta} \cdot r_2 e^{-i\theta}$ is real and positive (polar form, `cos²+sin²=1`).
- **1D diamagnetic inequality**: $a \leq \|a + bi\|$ (from $\|z\|^2 = a^2 + b^2 \geq a^2$).
- **Lattice Green's function**: $\|G(n)\| \leq 1/m^2$ (12 theorems from Mathlib Fourier API on `ZMod`).
- **Continuum limit**: OS0–OS2 for the O(N) LSM on T_L^2.

14.3 What is axiomatized (21 axioms)

Axiom	Content	Difficulty
`contour_deformation`	HS + Cauchy + QHJ + FK $\to |C| \leq M^{-1}$	research
`green_exponential_decay`	$M^{-1} \leq Ce^{-m_0|x|}$ (Fourier)	medium
`resolvent_complex_bound`	$\|(M+iV)^{-1}\| \leq M^{-1}$ (FK)	medium
`quantum_thimble_exists`	Φ exists with BL variance	hard
6 diamagnetic sub-axioms	heat kernel, Trotter, Laplace transform	medium
4 continuum limit	porting from pphi2/gaussian-field	routine
3 matrix calculus	det C^∞, Jacobi, Hessian	routine
2 contour shift	Cauchy for rectangles (Mathlib)	easy
1 Green's decay (sharp)	exponential rate on torus	medium

14.4 Open questions

1. **Decomposing `contour_deformation`**: This single axiom packages the entire bridge from the O(N) measure to the thimble analysis. Breaking it into sub-axioms (HS formula, Cauchy shift, QHJ existence, FK domination, triangle inequality) would make each piece independently verifiable.

2. **Singularity avoidance**: Can we prove $\mathcal{J}_0$ avoids $\det(-\Delta + 2i\sigma z) = 0$? The zeros are purely imaginary ($\sigma = ik^2/(2z)$), at distance $\geq m_0^2/(2z) = m_0^2\sqrt{N}/2$ from the saddle. For large N this should be straightforward.

3. **Global log-concavity**: Is $\operatorname{Re} f$ concave on all of $\mathcal{J}_0$, or only near the saddle? The $\operatorname{Tr}\log$ term introduces non-convexity far from the saddle. For BL: we need convexity within the $O(1/\sqrt{N})$-width peak, which holds by the Hessian computation.

4. **Intersection numbers**: Can we verify $n_0 = 1$? For the dominant saddle on the real axis with f'' positive definite, homotopy arguments should give $n_0 = 1$.

5. **Threshold N_0**: What is the minimal N for single-thimble dominance? Kupiainen's threshold for Strategy A is $N > C\, m_0^{-25}$ (very large). Can Strategy D improve this?

6. $N = 1$ **test case**: In $d = 0$ (one site), $\mathcal{J}_0$ is a curve in $\mathbb{C}$ that can be computed explicitly. Does the thimble analysis reproduce the known $\mathrm{P}(\varphi)_2$ mass gap?

7. **Flowed manifold vs. exact thimble**: Is the flowed manifold $\mathcal{M}_T$ for finite T sufficient for the proof, avoiding the need to characterize $\mathcal{J}_0$ globally?

14.5 Gap between this document and the Lean formalization

The Lean development mirrors the top-level proof architecture (Sections 7–10) but does not yet formalize the detailed mathematics:

- The thimble geometry (Sections 8–8.8), Hamilton-Jacobi theory (Section 9), and solution theory (Section 10) are documented but axiomatized in Lean. The `quantum_thimble_exists` axiom packages the IFT existence + BL variance bound.
- The "trivial averaging" argument (step 3 of the proof plan) is not separately formalized; Lean packages the entire bridge (HS + Cauchy + QHJ + FK + averaging) into the single axiom `contour_deformation`.
- The diamagnetic inequality has a proof structure (`DiagmagneticInequality.lean`, 413 lines) but the sub-steps (Trotter, heat kernel, Laplace transform) remain axioms.
- The Green's function Fourier analysis (`GreenDecay.lean`, 291 lines) has 12 proved theorems but the sharp exponential rate is axiomatized.

The Lean development is a **coherent, compiling scaffold** for the proof strategy described in this document, not yet a complete formal derivation of the analytic steps.

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